Tasks

DAX Calculations

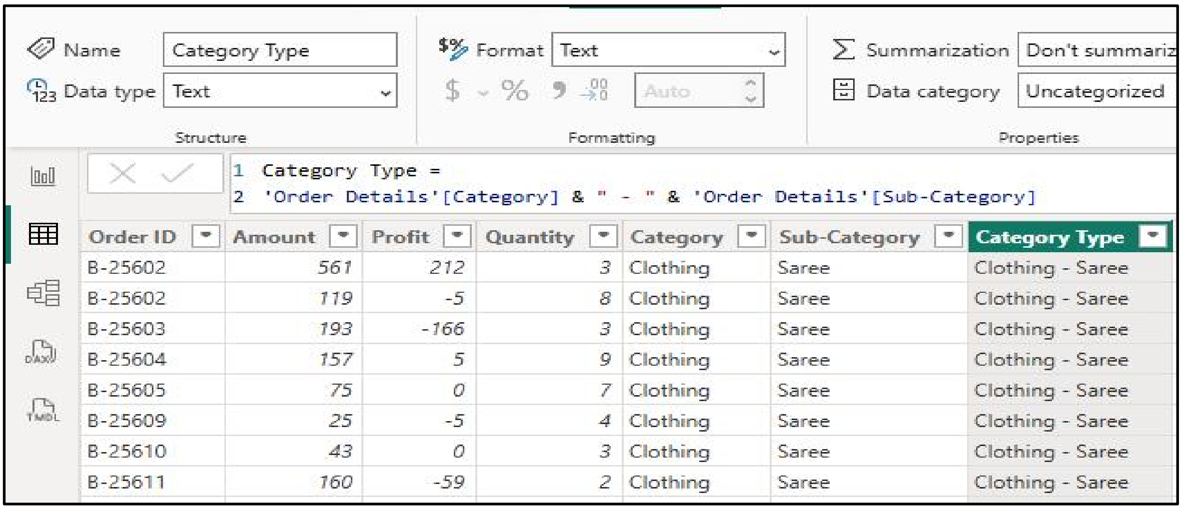
Calculated Columns

● Create a Calculated Column for 'Category Type':

Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Formula :

Category Type =
'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]




```
1 Category Type =
2 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]
```

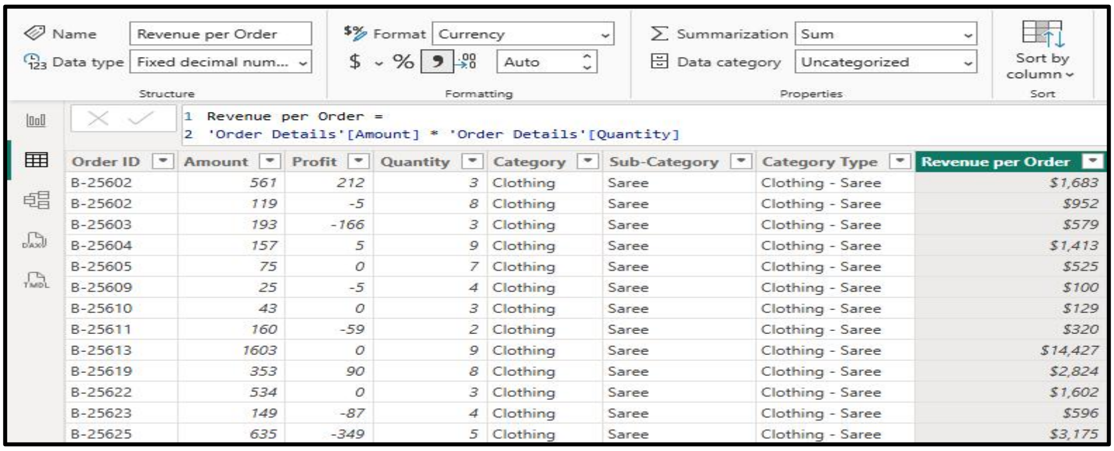
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25602	561	212	3	Clothing	Saree	Clothing - Saree
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree
B-25604	157	5	9	Clothing	Saree	Clothing - Saree
B-25605	75	0	7	Clothing	Saree	Clothing - Saree
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree
B-25610	43	0	3	Clothing	Saree	Clothing - Saree
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree

● Calculate Revenue per Order in Order Details Table:

Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

Formula :

Revenue Per Order =
'Order Details'[Amount] * 'Order Details'[Quantity]




```
1 Revenue per Order =
2 'Order Details'[Amount] * 'Order Details'[Quantity]
```

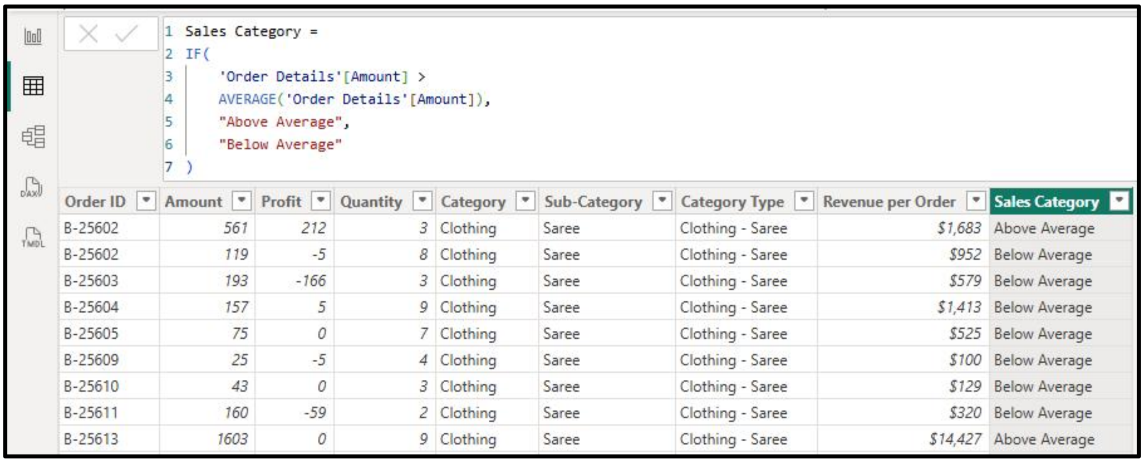
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	$1,683
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	$952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	$579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	$1,413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	$525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	$100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	$129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	$320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	$14,427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	$2,824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	$1,602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	$596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	$3,175

- **Create a Calculated Column to Categorize Sales:**

Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Formula :

```
Sales Category =
IF(
    'Order Details'[Amount] >
    AVERAGE('Order Details'[Amount]),
    "Above Average",
    "Below Average"
)
```



```
Sales Category =
IF(
    'Order Details'[Amount] >
    AVERAGE('Order Details'[Amount]),
    "Above Average",
    "Below Average"
)
```

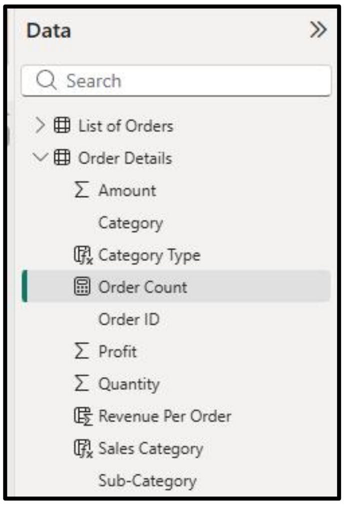
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	$1,683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	$952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	$579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	$1,413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	$525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	$100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	$129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	$320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	$14,427	Above Average

Calculated Measures

1. **Order Count-** Create a measure that calculates the **total number of orders** in the **Order Details** table.

Formula :

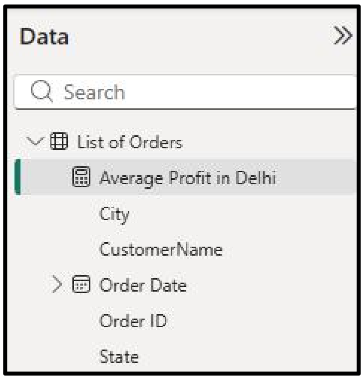
```
Order Count = COUNT('Order Details'[Order ID])
```



2. **Average Profit in Delhi-** Create a measure that calculates the **average profit for orders placed in Delhi**.

Formula:

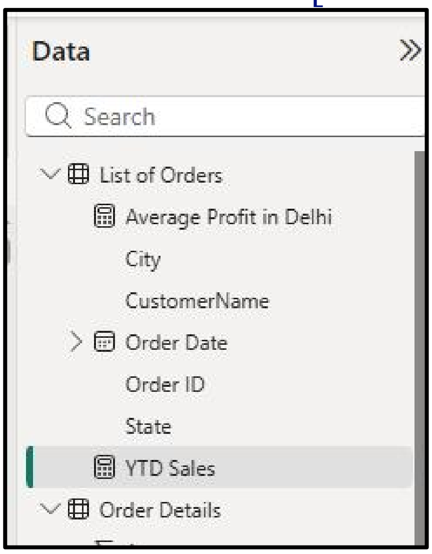
```
Average Profit in Delhi =
CALCULATE(
    AVERAGE('Order Details'[Profit]),
    'List of Orders'[City] = "Delhi"
)
```



3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date**.

Formula:

```
YTD Sales =
TOTALYTD(
    SUM('Order Details'[Amount]),
     'List of Orders'[Order Date])
```



Data Visualization

Create the following visualizations using Power BI:

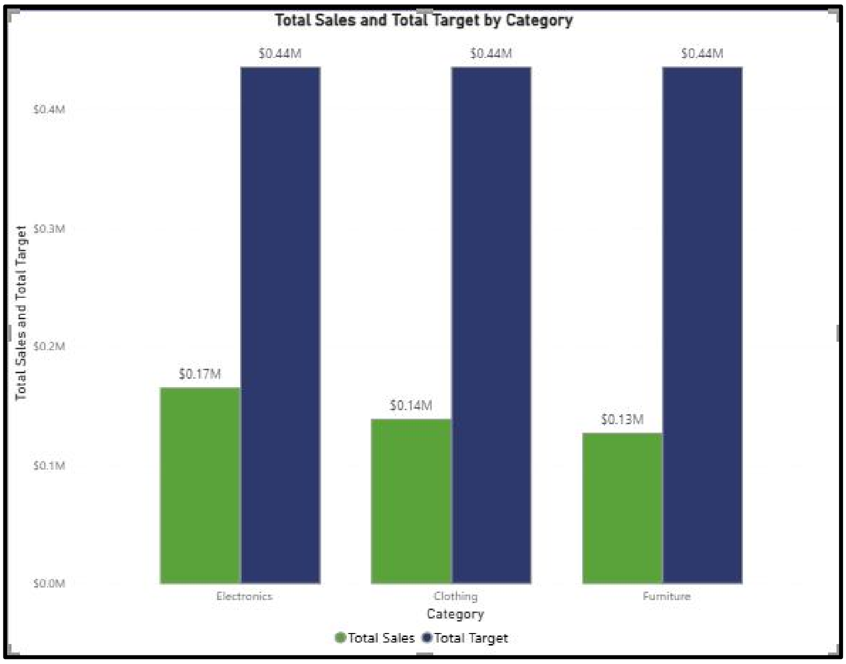
● Sales Target Achievement by Category:

Compare actual sales with sales targets by category using a clustered column chart.

Visual: Clustered Column Chart

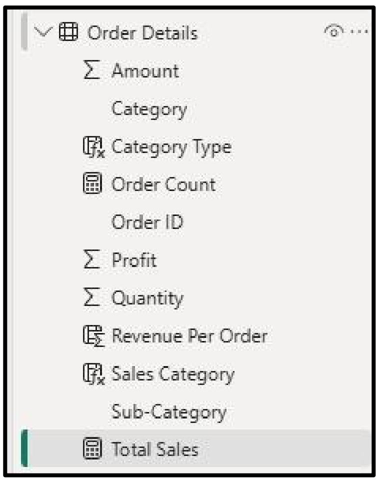
Fields

- **Axis:** Category
- **Values:** Total Sales
- **Values:** Sales Target

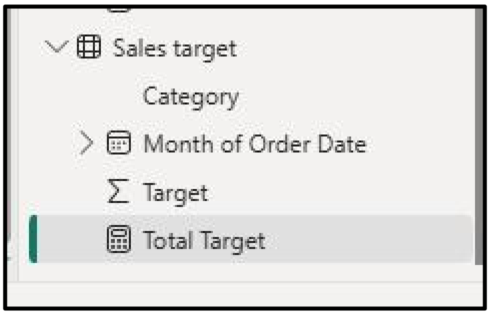


Measures:

Total Sales =SUM('Order Details'[Amount])



Total Target =
SUM('Sales Target'[Target])



Purpose:

- Compare actual vs target.
- Identify categories exceeding targets.

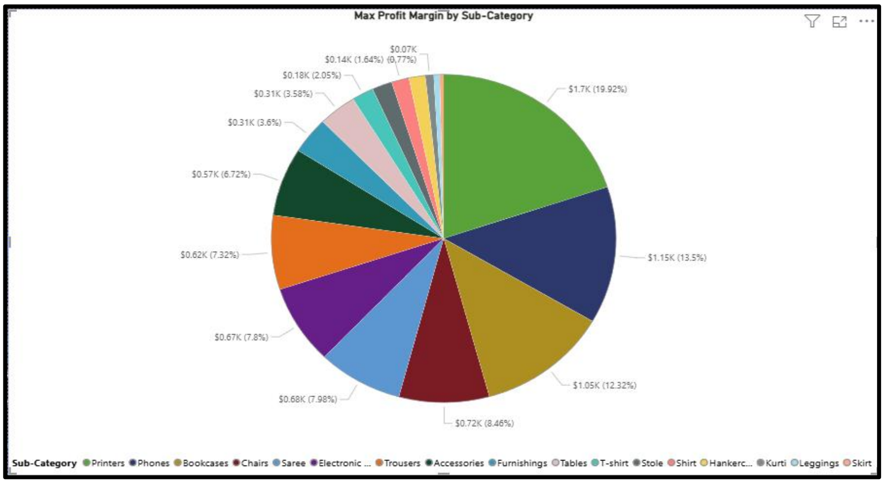
● Max Profit Margin by Sub-Category:

Analyze the maximum profit margin for each sub-category of products using a donut chart.

Visual: Donut Chart

Fields

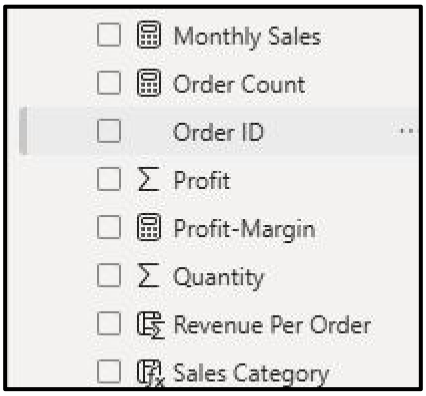
- Legend → Sub-Category
- Values → Max Profit Margin

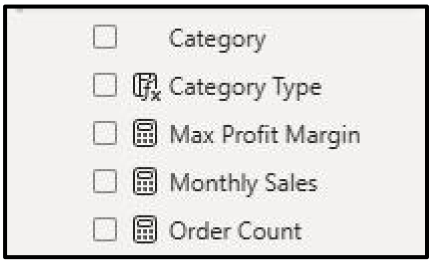


Create Measure:

Profit-Margin =
DIVIDE(
    SUM('Order Details'[Profit]),
    SUM('Order Details'[Amount]),

0
)
Max Profit Margin =
(
MAX('Order Details'[Profit])
)





Purpose:

- Find highest margin product groups.

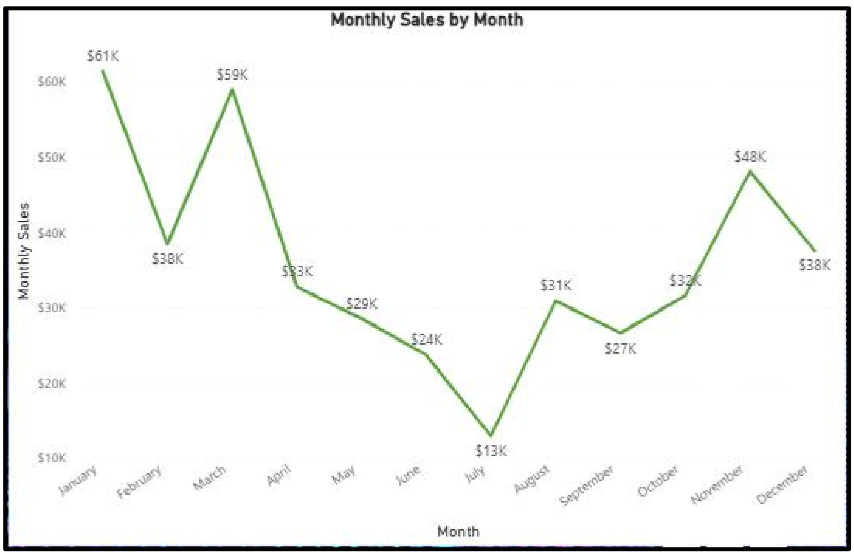
● Monthly Sales Trend:

Show the trend of monthly sales over time using a line chart.

Visual: Line Chart

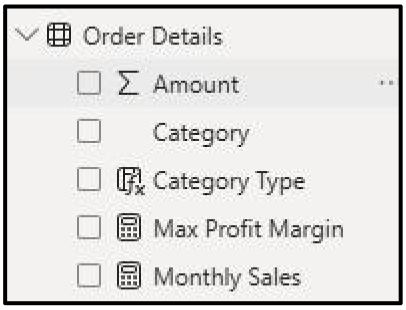
Fields

- X-Axis → Order Date (Month)
- Y-Axis → Total Sales



Measure:

Monthly Sales =
SUM('Order Details'[Amount])



Purpose:

- Observe sales growth and seasonality.

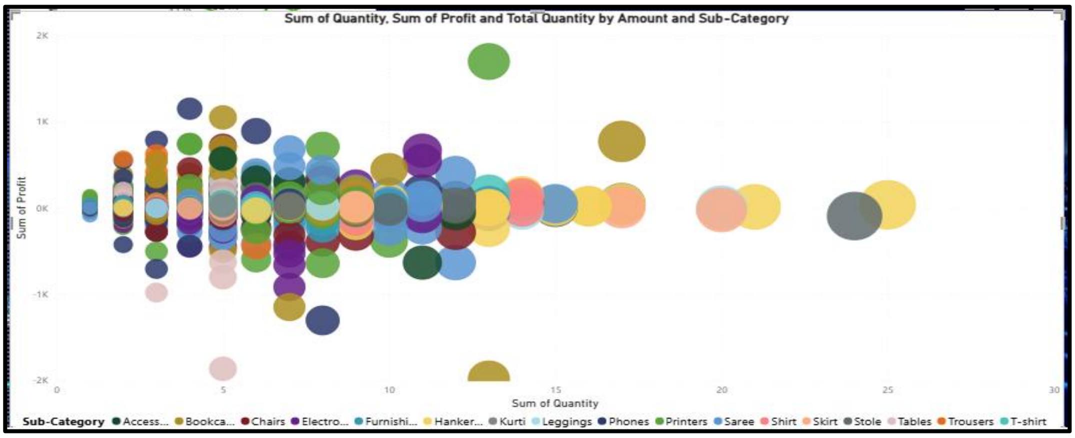
● Comparison of Profit and Quantity by Sub-Category:

Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

Visual: Scatter Chart

Fields

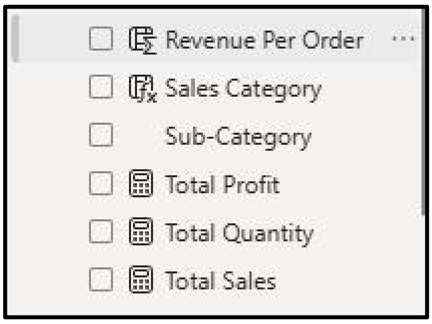
- X Axis → Quantity
- Y Axis → Profit
- Details → Sub-Category
- Size → Sales



Measures:

```
Total Quantity =
SUM('Order Details'[Quantity])

Total-Profit =
SUM('Order Details'[Profit])
```



Purpose:

- Identify high-profit/high-volume products.

● Comparison of Total Sales Amount and Target:

Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

Visual: Cards

Create:

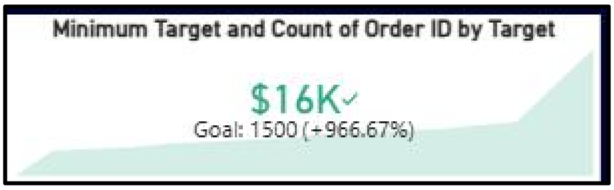
Total Sales Amount =
SUM('Order Details'[Amount])

Sales Target =
SUM('Sales Target'[Target])

Add:

- Card → Total Sales Amount
- Card → Sales Target

● **Multi-row Card (Minimum Target by Segment)**



Measure:

Minimum Target =
MIN('Sales Target'[Target])

Fields:

- Segment
- Minimum Target

Purpose:

- Quick target comparison.

● **Sales Performance Matrix:**
Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

Visual: Matrix

Rows:

- Category

Columns:

- Month

Values:

- Total Sales
- Sales Target

Quantity	Clothing	Electronics	Furniture	Total
1	$174,000	$129,000	$132,900	$435,900
2	$174,000	$129,000	$132,900	$435,900
3	$174,000	$129,000	$132,900	$435,900
4	$174,000	$129,000	$132,900	$435,900
5	$174,000	$129,000	$132,900	$435,900
6	$174,000	$129,000	$132,900	$435,900
7	$174,000	$129,000	$132,900	$435,900
8	$174,000	$129,000	$132,900	$435,900
Total	$174,000	$129,000	$132,900	$435,900

Optional:

Target Achievement % =
DIVIDE(
  'Order Details'[Total Sales],
  'Sales target'[Total Target],
  0
)

Purpose:

- Month-by-month performance tracking.

● Geographic Sales Analysis:
Visualize total sales on a map by city to identify regional sales patterns.

Visual: Map

Fields:

- Location → City
- Size → Total Sales

Measure:

City Sales =
SUM('Order Details'[Amount])

Purpose:

- Identify strongest sales regions.

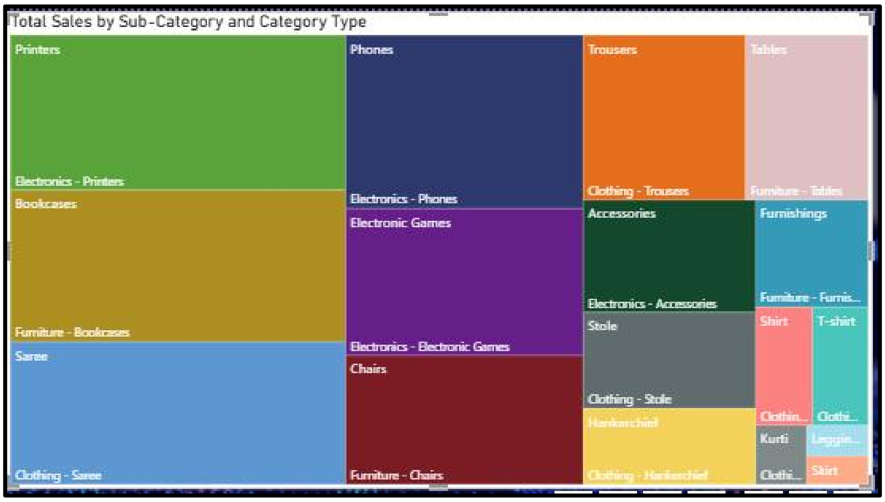
● Sales Distribution by Sub-Category:

Represent the sales distribution across different
● sub-categories using a tree map.

Visual: Treemap

Fields:

- Group → Sub-Category
- Values → Total Sales



Measure:

Sub Category Sales =
SUM('Order Details'[Amount])

Purpose:

- Understand sales contribution.

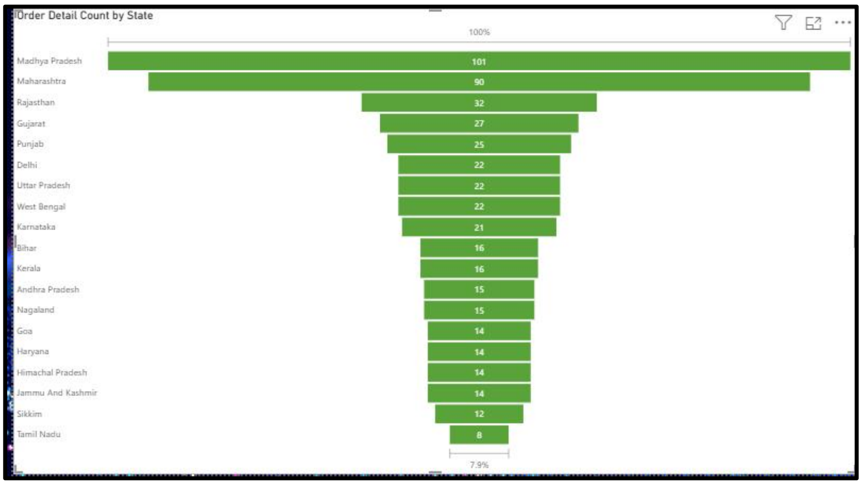
● Order Count Analysis by State:

Create a funnel chart to visualize the distribution of order counts across different states.

Visual: Funnel Chart

Fields:

- Group → State
- Values → Order Count



Measure:

Order Count =
DISTINCTCOUNT('Order Details'[Order ID])

Purpose:

- Compare order volume across states.

● DashBoard